ECO392: Shiv Nadar Institution of Eminence

Doomed By The Rains?

A climate based poverty profile of Tanzania

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Group no.: **7**

Tanzanian Context

- Stable political landscape but uneven regional development
- Majority rural; livelihoods depend on rain-fed agriculture
- Strong climate variation → different exposure to shocks
- Ideal setting to study how climate + infrastructure shape poverty

Why Adult-Equivalence Consumption?

Formula Used

- We use the **OECD-modified adult-equivalence scale**:
  1.0 for the first adult, **0.7** for additional adults, **0.5** for each child.
- This converts household expenditure into **adult-equivalent consumption**.

Why We Used It

- Raw per-household spending **overstates welfare** for large families.
- Adult-equivalence adjusts for household size & composition → giving a **fairer and more comparable measure of welfare** across regions and rainfall groups.

Why These Region Subgroups?

Two factors shape livelihood risk:

1. Rainfall quantity (Wet / Dry)

2. Rainfall stability (Stable / Unstable)

Stability affects planning, buffer stocks, market access, and vulnerability to crop failures.

This creates **four climate regimes**:
| Wet+Stable | Wet+Unstable | Dry+Stable | Dry+Unstable | a clean climate-risk classification (used widely in climate impact literature).

WET + STABLE	DRY + UNSTABLE
LINDI	ARUSHA
MARA	DODOMA
MTWARA	KILIMANJARO
MWANZA	MANYARA
PEMBA NORTH	MBEYA
RUVUMA	MOROGORO
SINGIDA	NJOMBE
DRY + STABLE	**WET + UNSTABLE**
DAR ES SALAAM	GEITA
PEMBA SOUTH	IRINGA
PWANI	KAGERA
RUKWA	KATAVI
SIMIYU	KIGOMA
SONGWE	SHINYANGA

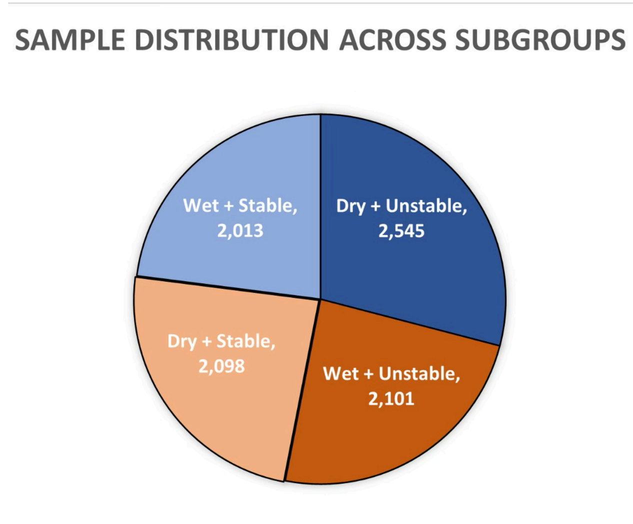


FIGURE 1: *Sample split*

Descriptive Stats: What the Data Show

- Total sample: 8,757 households, well-balanced across 4 climate groups.
- Welfare is geographically clustered even before poverty analysis.
- Dry+Stable areas show better initial welfare than expected; Wet+Unstable surprisingly lower.

FIGURE 2: *Mean adult-equivalent consumption*

-> region_group = Dry+Stable

Variable	Obs	Weight	Mean
exp_ae_oecd	2,098	2341170.46	222661.2

-> region_group = Wet+Stable

Variable	Obs	Weight	Mean
exp_ae_oecd	2,013	2458427.32	112155.1

-> region_group = Dry+Unstable

Variable	Obs	Weight	Mean
exp_ae_oecd	2,545	3095952.62	140336.6

-> region_group = Wet+Unstable

Variable	Obs	Weight	Mean
exp_ae_oecd	2,101	2061491.15	96094.28

National Poverty Levels (FGT)

Using weighted consumption and BNPL:

- Tanzania Mainland shows higher **headcount** than **depth & severity** → with a lower squared gap than headcount, the average poor person is more similar to another poor person.

FIGURE 3: *National FGT Indices*

Foster-Greer-Thorbecke poverty indices, FGT(a)

All obs	a=0	a=1	a=2
	0.22234	0.06457	0.02666

Poverty Profiles Across Subgroups

- **Wet+Unstable** emerges as the most vulnerable climate regime.
- **Dry+Stable** shows lowest depth & severity, stability is protective.
- **Dry+Unstable** performs better than average, consistent with climate-risk theory.
- As expected, dry regions have a lower headcount and squared gap.

FIGURE 4: *FGT0 / FGT1 / FGT2 by Region Group*

Subgroup FGT index estimates, FGT(a)

region_group	a=0	a=1	a=2
Wet+Stable	0.26031	0.07880	0.03316
Wet+Unstable	0.34032	0.10626	0.04557
Dry+Stable	0.15662	0.04320	0.01744
Dry+Unstable	0.16659	0.04542	0.01800

Why Do These Patterns Occur? (Hypothesis + Evidence)

Why Wet+Unstable performs worst

- High rainfall variability → repeated crop loss
- Remoteness & flood-prone terrain → poor market access
- Shock-sensitive crop mix
- Climate literature: instability destroys assets & depresses productivity
  Conclusion: Instability is the binding constraint.

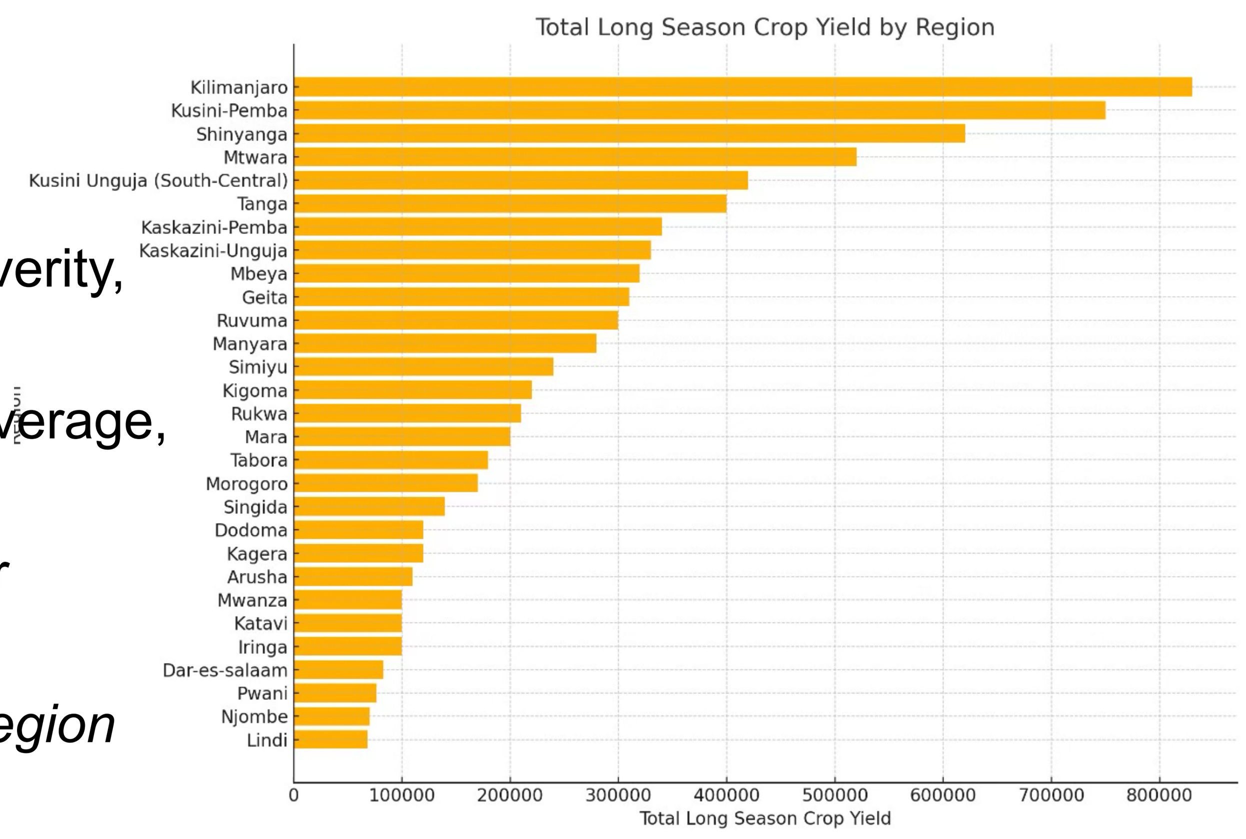
Why Dry+Stable performs best

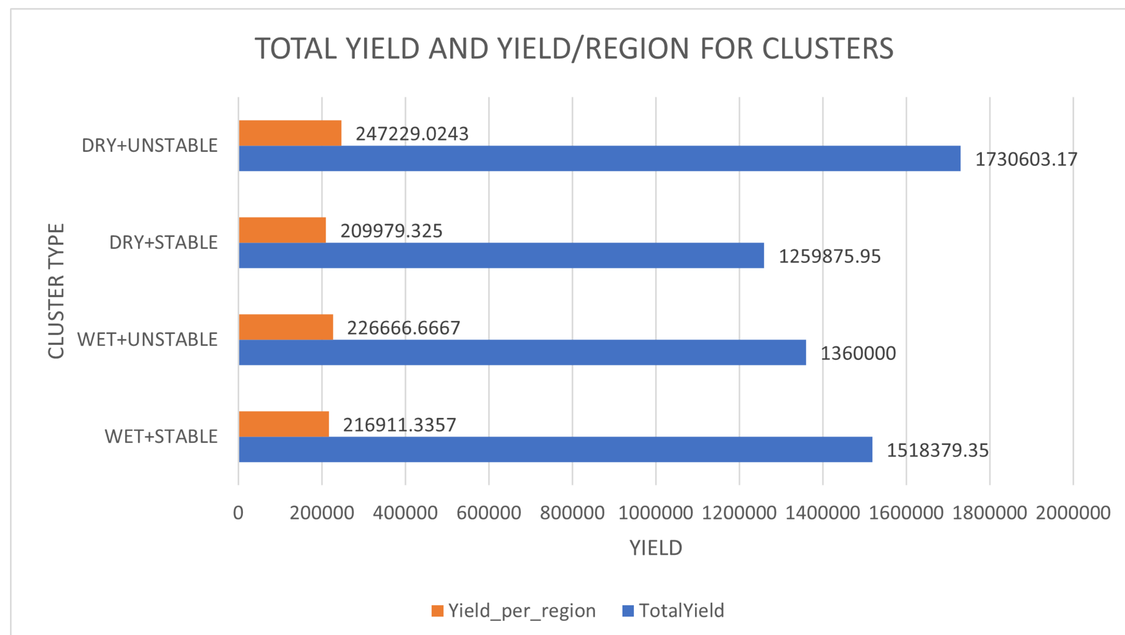
- Predictable climate → better planning, fewer losses
- Many peri-urban/urban-adjacent zones → diversified income
- Higher access to markets & services
  Conclusion: Stability + infrastructure reduce vulnerability.

Agriculture Analysis (Rainy-Season Crops Only)

- Wet+Stable regions have **the second-highest yields** yet show **higher poverty than expected**.
- In an agriculture-dependent economy, we would expect Wet+Stable to have **the lowest poverty**, but this pattern does not appear.
- This suggests that **yields alone don't translate into welfare,** factors like **market access, infrastructure, and diversification** shape outcomes beyond climate.

FIGURE 5: *Long-Season Crop Yield by Region*





Policy Directions

- Prioritise **Wet+Unstable** and **Dry+Stable** regions in adaptation planning.
- Promote **irrigation, crop diversification, and climate-resilient varieties**.
- Strengthen **transport access** to reduce market vulnerability.
- Expand **TASAF** and shock-responsive safety nets in wet climate zones.
- Invest in **local early-warning systems** + farmer-level advisories.

Sources

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